INSTITUT UNIVERSITAIRE DES SCIENCES

Faculté des Sciences et technologies

TD N°3 – Réseaux I

Adressage IPv4 et IPv6 avec Cisco Packet Tracer

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1. Description des résultats de la tâche

1.1 Objectif du TD

Ce TD a pour but de se familiariser avec l'adressage IP et la configuration réseau dans le simulateur Cisco Packet Tracer. Il s'agissait de :

- attribuer des adresses IP valides aux machines d'un réseau local ;
- comprendre l'adressage IPv4 et IPv6 ;
- configurer des adresses IP sur des hôtes (PC, laptop, serveur) dans Cisco Packet Tracer ;
- vérifier la connectivité avec la commande ping (IPv4) et ping vers une adresse IPv6.

1.2 Notions utilisées

- **IPv4, ipv6, hub ,switch, ping**

1.3 Démarche suivie

1. Placer les équipements dans Packet Traceret les relier avec des câbles droits
2. Sur chaque appareil : double-clic, onglet Desktop puis IP Configuration (pour le serveur : onglet Config puis FastEthernet0), choix du mode Static, saisie de l'adresse IPv4 avec son masque et de l'adresse IPv6 avec sa longueur de préfixe.
3. Laisser la passerelle par défaut et le DNS vides (0.0.0.0) : le réseau est unique et ne comporte aucun routeur.
4. Ouvrir Desktop puis Command Prompt et lancer ping <adresse> vers les autres machines.
5. Relever le résultat (paquets envoyés, reçus, perdus) et réaliser une capture d'écran de chaque étape.
6. Rédiger le rapport, l'exporter en PDF et l'héberger sur GitHub avec le dossier des captures.

Remarque sur l'IPv6. Le préfixe « 3000:is9:1::/64 » de l'énoncé n'est pas une adresse IPv6 valide : les lettres « i » et « s » ne sont pas des chiffres hexadécimaux, et Packet Tracer refuse une telle saisie. Nous avons donc utilisé le préfixe 3000:159:1::/64, qui conserve la même structure.

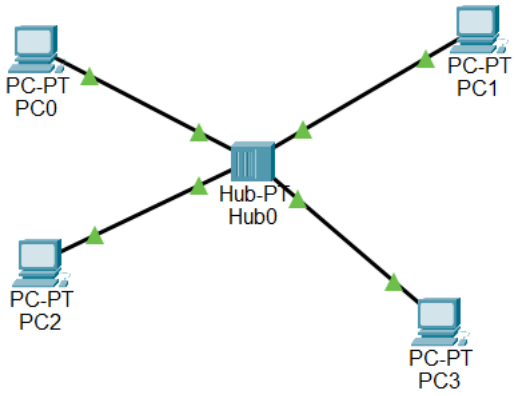
Correspondance avec les topologies. Le tableau compte huit appareils. Les appareils sont affectés dans l'ordre : Client1 à Client6, puis Admin, puis Server Mail.

Dispositif	IPv4	Masque	IPv6 (utilisée)
client 1	192.168.1.10	255.255.255.0	3000:159:1::1 /64
client 2	192.168.1.11	255.255.255.0	3000:159:1::2 /64
client 3	192.168.1.12	255.255.255.0	3000:159:1::3 /64
client 4	192.168.1.13	255.255.255.0	3000:159:1::4 /64
client 5	192.168.1.14	255.255.255.0	3000:159:1::5 /64
client 6	192.168.1.15	255.255.255.0	3000:159:1::6 /64
client 7	192.168.1.16	255.255.255.0	3000:159:1::7 /64
client 8	192.168.1.17	255.255.255.0	3000:159:1::8 /64

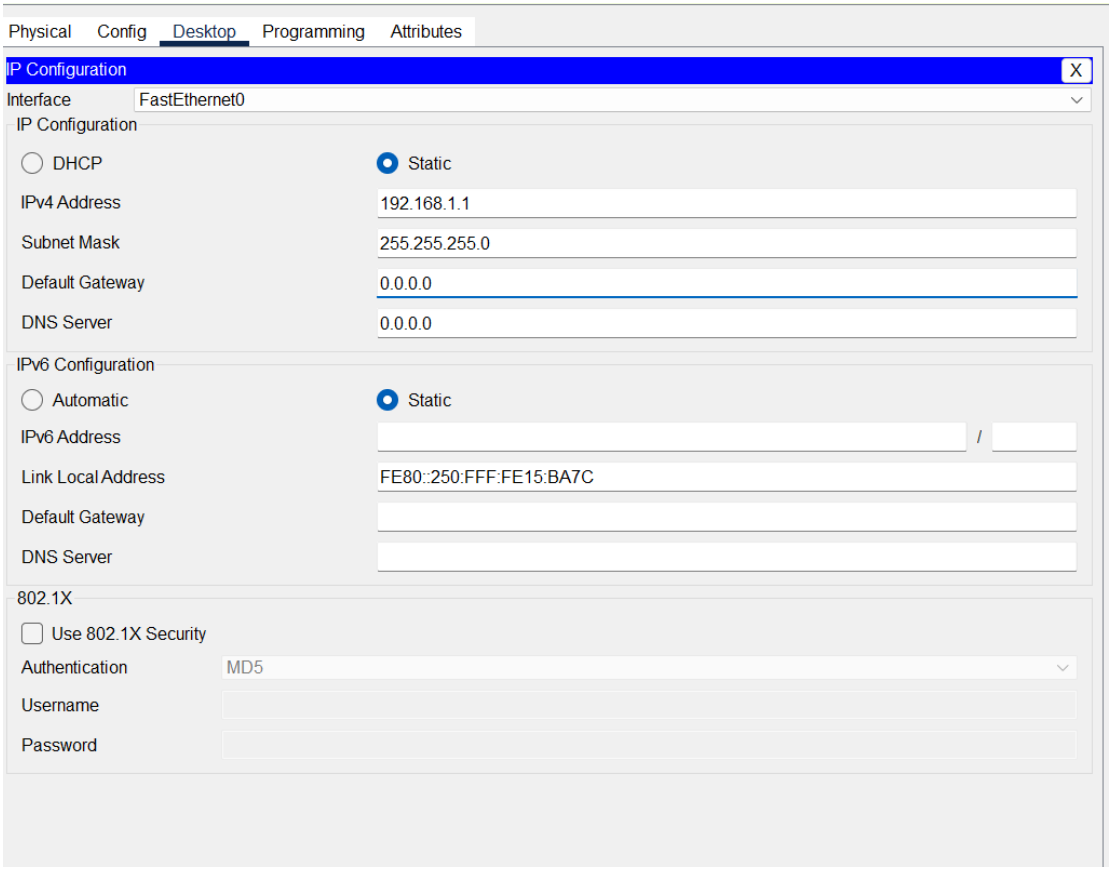
2. Résultats de l’exécution des commandes

2.1 Partie guidée – IPv4 (hub et 4 PC)

Topologie : un hub relié à quatre PC. Adressage appliqué :



Les quatre PC sont reliés au hub par des câbles droits. Les liaisons sont actives (triangles verts) : le réseau est prêt à être configuré.



L’adresse 192.168.1.1 et le masque 255.255.255.0 sont saisis en mode Static. Les trois autres PC sont configurés de la même façon avec 192.168.1.2, 192.168.1.3 et 192.168.1.4.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time=3ms TTL=128
Reply from 192.168.1.1: bytes=32 time=3ms TTL=128
Reply from 192.168.1.1: bytes=32 time=3ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 3ms, Average = 2ms

C:\>
```

```
C:\> ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=4ms TTL=128
Reply from 192.168.1.2: bytes=32 time=4ms TTL=128
Reply from 192.168.1.2: bytes=32 time=2ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 4ms, Average = 2ms
```

```
C:\> ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time=1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

```
C:\> ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

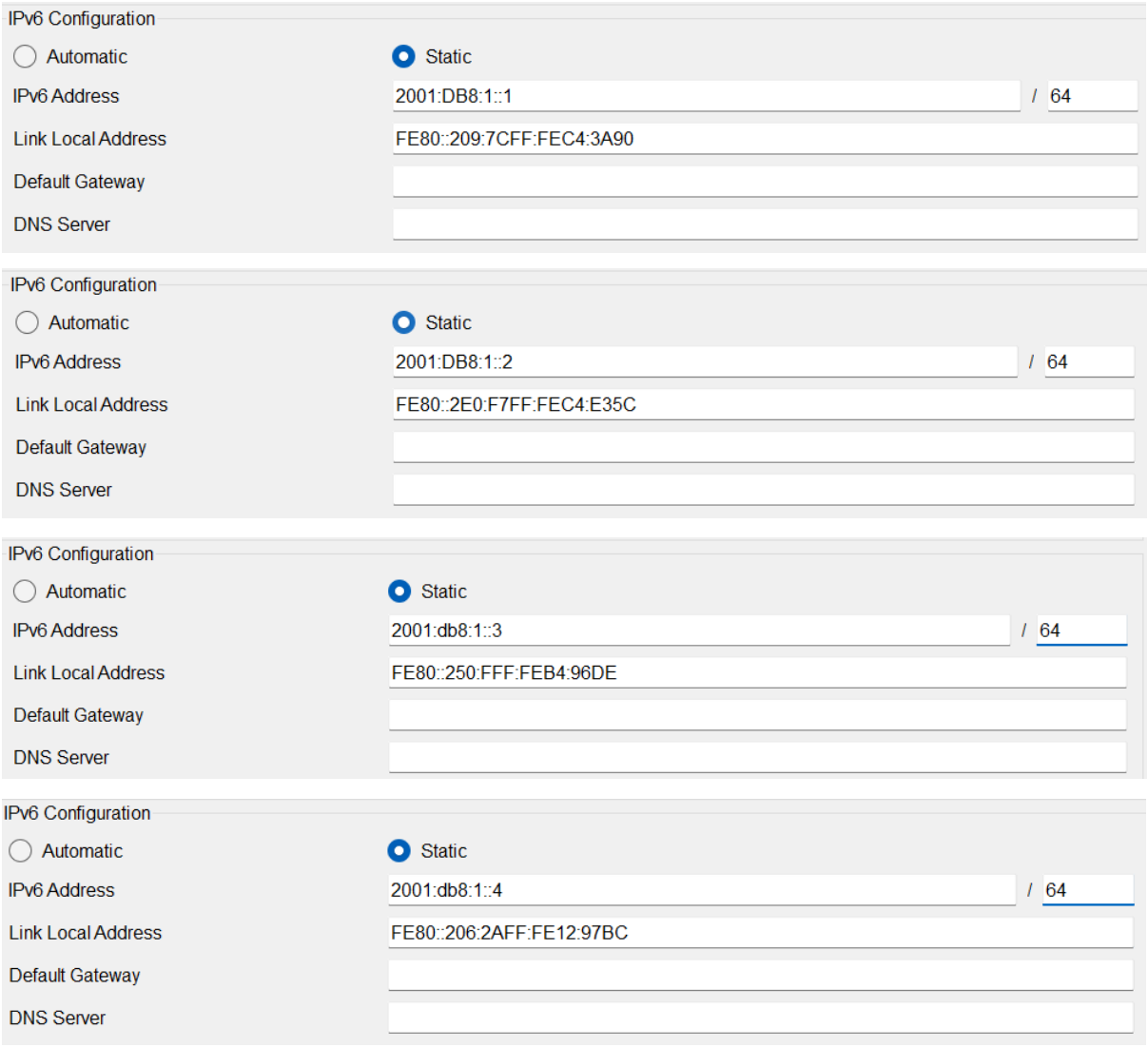
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

La commande ping 192.168.1.1 envoie 4 paquets de 32 octets. Les 4 réponses reçues (0 % de perte) prouvent que PC1 et PC2 communiquent. Le même test est répété entre tous les PC.

2.2 Partie guidée – IPv6

Dans la même topologie, une adresse IPv6 est ajoutée sur chaque PC :



L'adresse 2001:db8:1::1 et le préfixe 64 sont saisis dans la zone IPv6 Configuration. L'adresse link-local (FE80::…) est générée automatiquement par Packet Tracer.

```
Reply from 2001:DB8:1::1: bytes=32 time=8ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time=1ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=128

Ping statistics for 2001:DB8:1::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 8ms, Average = 2ms
```

```
C:\>ping 2001:db8:1::2

Pinging 2001:db8:1::2 with 32 bytes of data:

Reply from 2001:DB8:1::2: bytes=32 time=1ms TTL=128
Reply from 2001:DB8:1::2: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::2: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::2: bytes=32 time<1ms TTL=128

Ping statistics for 2001:DB8:1::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

```
Pinging 2001:db8:1::3 with 32 bytes of data:

Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128

Ping statistics for 2001:DB8:1::3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>ping 2001:db8:1::4

Pinging 2001:db8:1::4 with 32 bytes of data:

Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::4: bytes=32 time=3ms TTL=128
Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128

Ping statistics for 2001:DB8:1::4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 3ms, Average = 0ms
```

La commande ping 2001:db8:1::1 reçoit 4 réponses sur 4. La connectivité IPv6 fonctionne donc dans le réseau 2001:db8:1::/64.

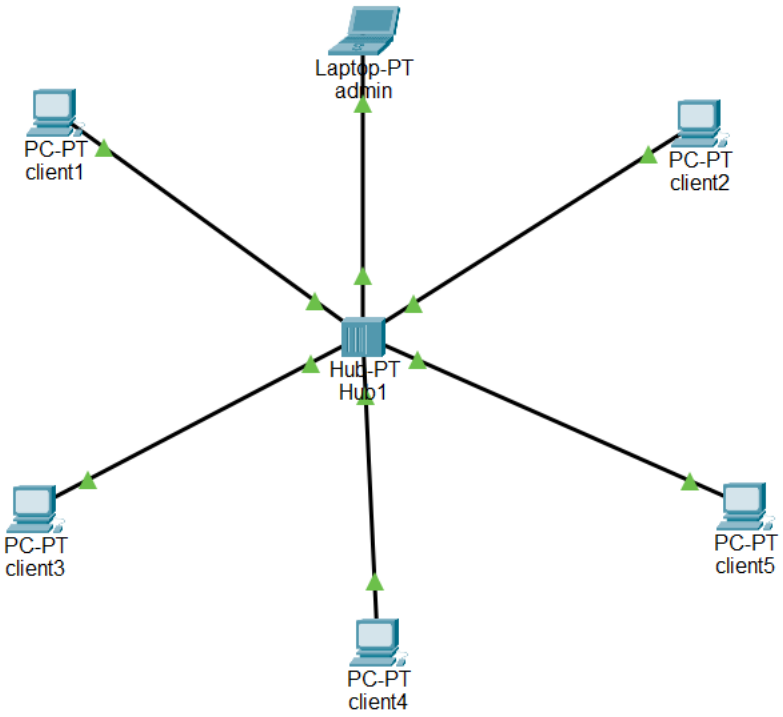
2.3 Travaux dirigés

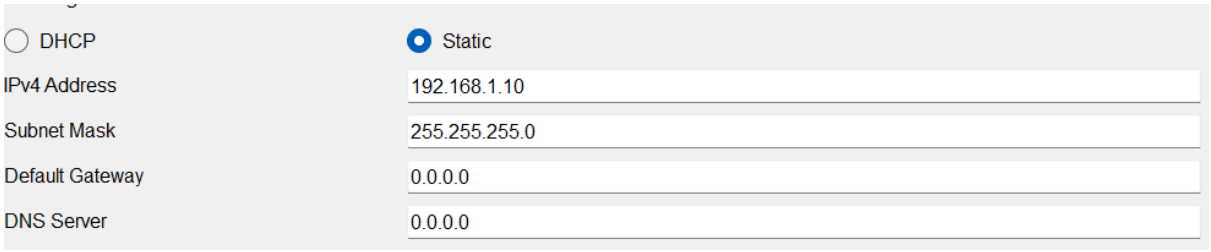
Exercice 1 – Topologie à hub, adresses IPv4

Le hub relie Client1 à Client5 et le laptop Admin.

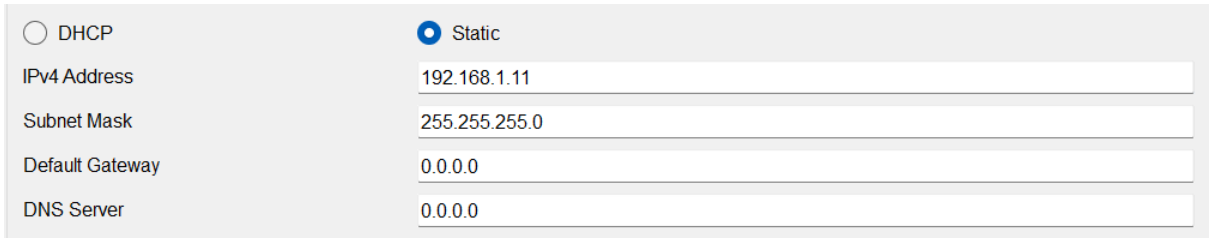
Appareil	Adresse IPv4	Masque
Client1	192.168.1.10	255.255.255.0
Client2	192.168.1.11	255.255.255.0
Client3	192.168.1.12	255.255.255.0
Client4	192.168.1.13	255.255.255.0
Client5	192.168.1.14	255.255.255.0
Admin (Laptop-PT)	192.168.1.15	255.255.255.0

Les six appareils sont connectés au même hub et configurés dans le réseau

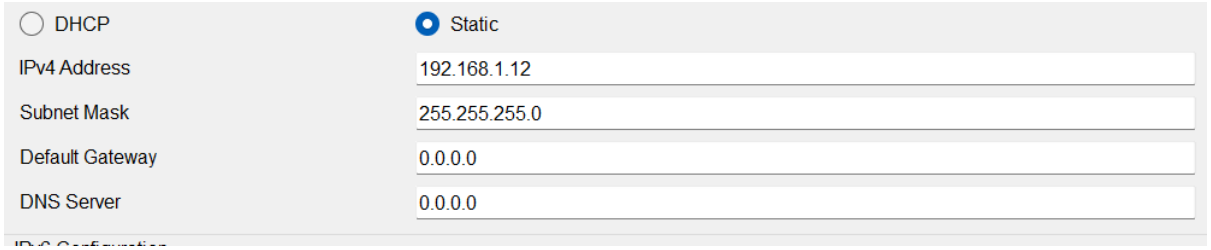




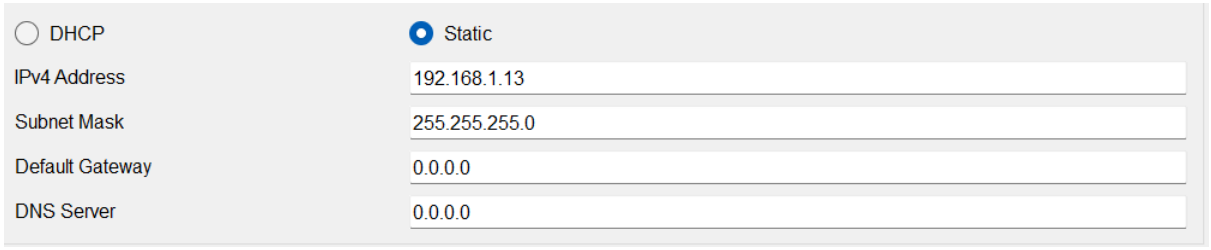
Client 1



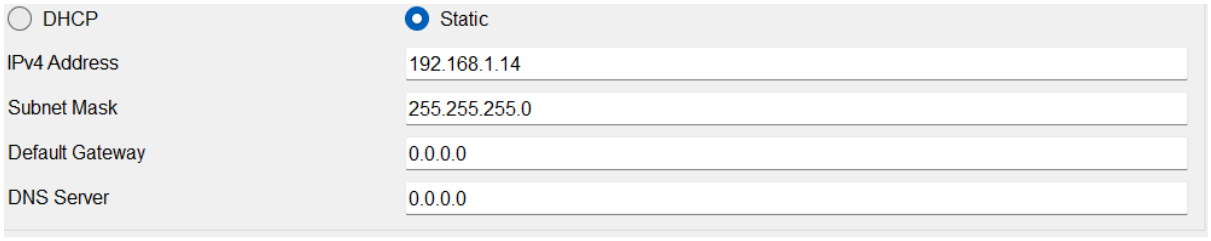
Client 2



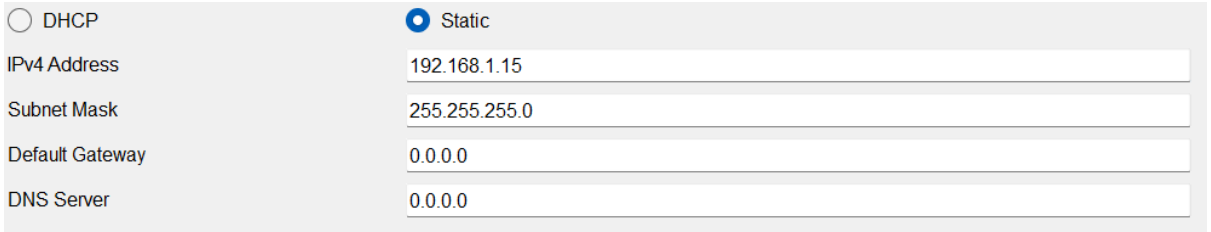
Client 3



Client 4



Client 5



Laptop-admin

```
C:\>ping 192.168.1.15

Pinging 192.168.1.15 with 32 bytes of data:

Reply from 192.168.1.15: bytes=32 time<1ms TTL=128
Reply from 192.168.1.15: bytes=32 time=5ms TTL=128
Reply from 192.168.1.15: bytes=32 time=6ms TTL=128
Reply from 192.168.1.15: bytes=32 time=2ms TTL=128

Ping statistics for 192.168.1.15:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 6ms, Average = 3ms
```

```
C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time=1ms TTL=128
Reply from 192.168.1.10: bytes=32 time=3ms TTL=128
Reply from 192.168.1.10: bytes=32 time=2ms TTL=128
Reply from 192.168.1.10: bytes=32 time=2ms TTL=128

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 3ms, Average = 2ms
```

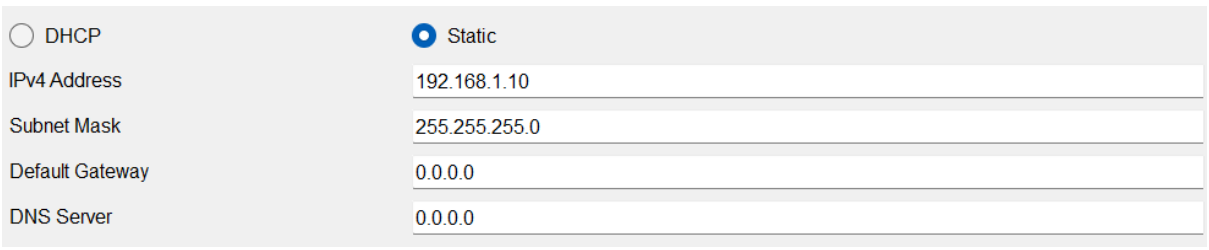
Admin reçoit une réponse de chacun des cinq clients. D'autres tests confirment que tous les appareils se joignent entre eux.

Exercice 2 – Topologie à switch, adresses IPv4

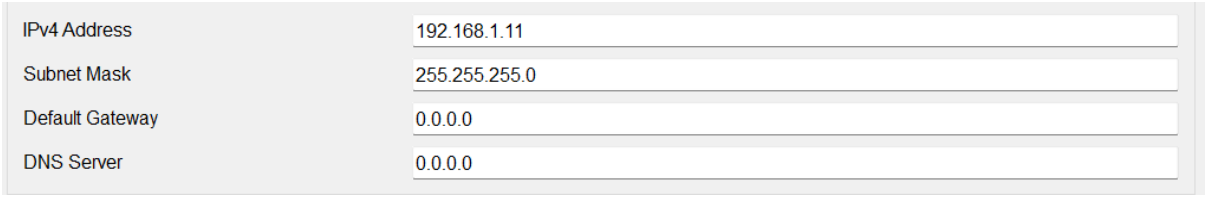
Le switch relie Client1 à Client6, le laptop Admin et le Server Mail.

Appareil	Adresse IPv4	Masque
Client1	192.168.1.10	255.255.255.0
Client2	192.168.1.11	255.255.255.0
Client3	192.168.1.12	255.255.255.0
Client4	192.168.1.13	255.255.255.0
Client5	192.168.1.14	255.255.255.0
Client6	192.168.1.15	255.255.255.0
Admin (Laptop-PT)	192.168.1.16	255.255.255.0
Server Mail (Server-PT)	192.168.1.17	255.255.255.0

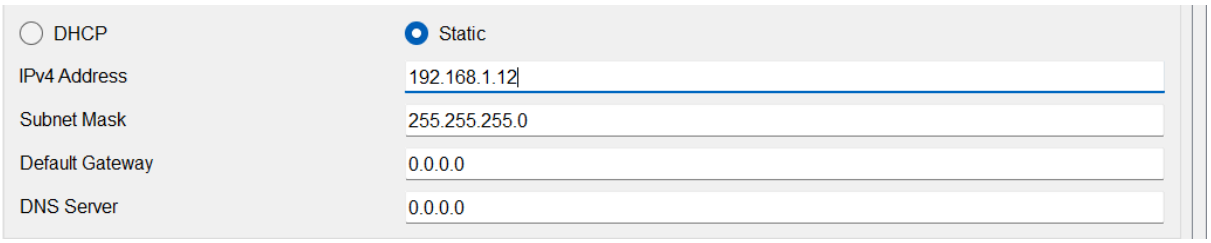
Les huit appareils sont reliés au switch et configurés dans le même réseau



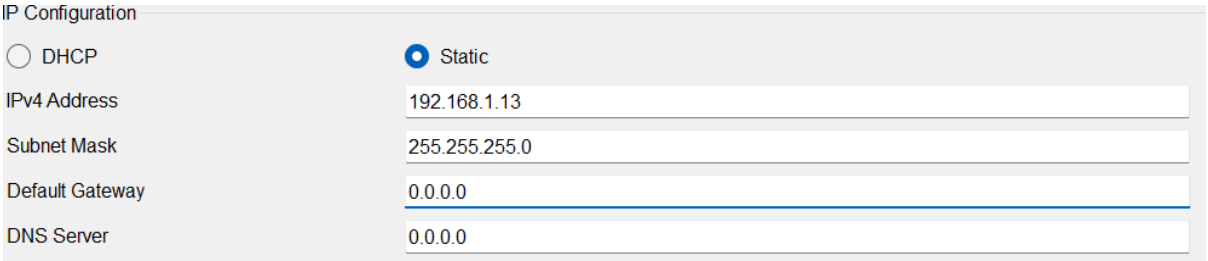
Client 1



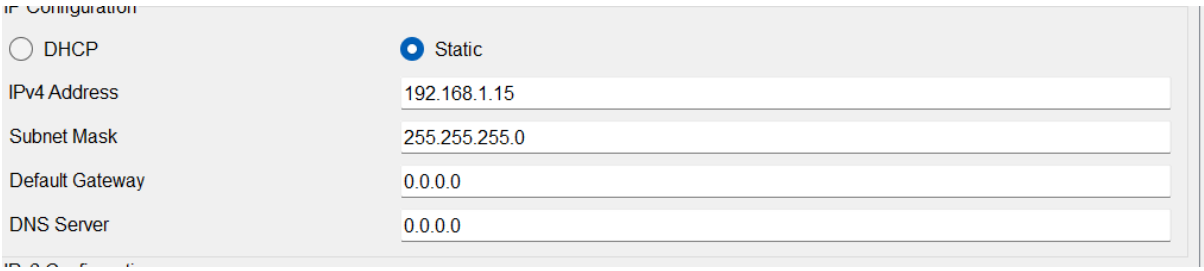
Client2



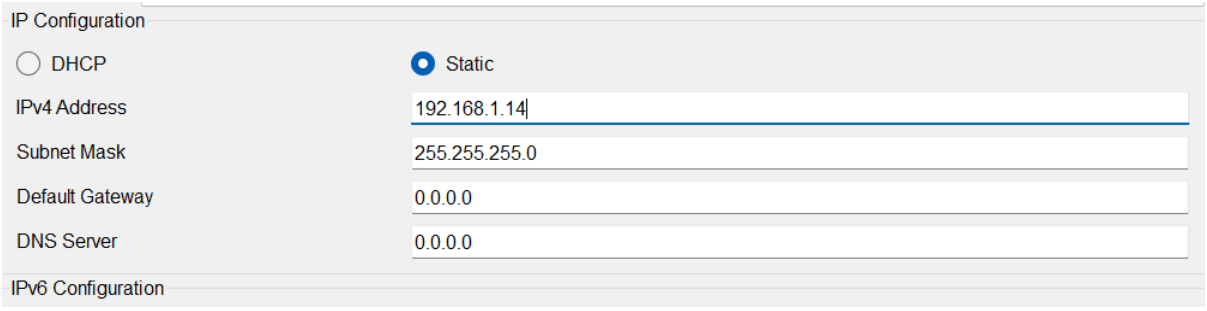
Client3



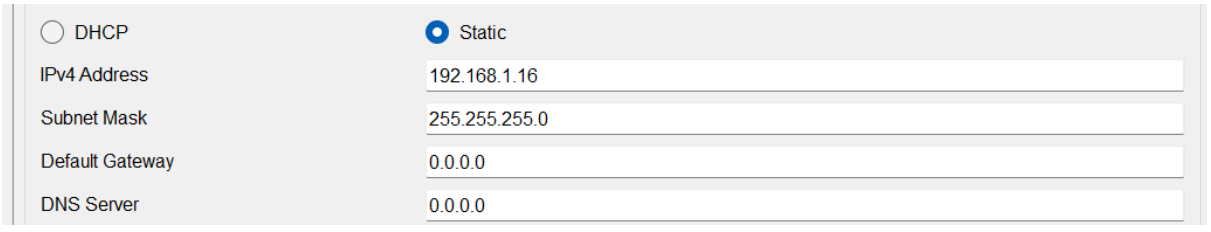
Client4



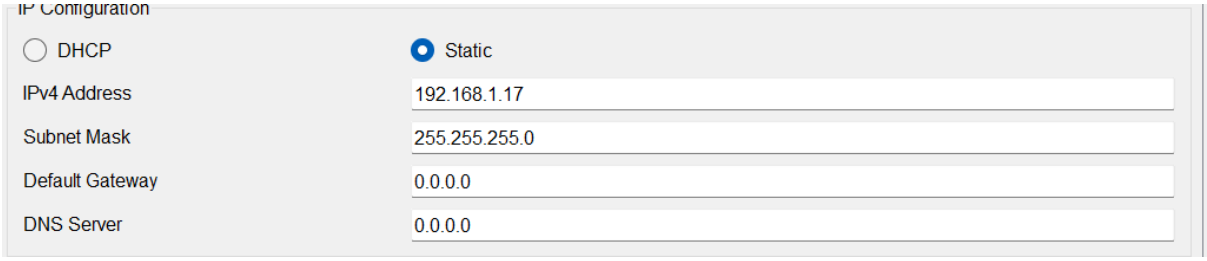
Client 6



Client5



Admin



Server

```
C:\>ping 192.168.1.17

Pinging 192.168.1.17 with 32 bytes of data:

Reply from 192.168.1.17: bytes=32 time<1ms TTL=128
Reply from 192.168.1.17: bytes=32 time<1ms TTL=128
Reply from 192.168.1.17: bytes=32 time<1ms TTL=128
Reply from 192.168.1.17: bytes=32 time=5ms TTL=128

Ping statistics for 192.168.1.17:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 5ms, Average = 1ms

C:\>
```

```
C:\>ping 192.168.1.16

Pinging 192.168.1.16 with 32 bytes of data:

Reply from 192.168.1.16: bytes=32 time<1ms TTL=128
Reply from 192.168.1.16: bytes=32 time<1ms TTL=128
Reply from 192.168.1.16: bytes=32 time<1ms TTL=128
Reply from 192.168.1.16: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.16:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Client1 obtient des réponses de Server Mail et d'Admin. Le switch transmet donc correctement le trafic entre tous les ports.

Exercice 3 – Choix d'une topologie : switch, IPv4

Nous avons choisi la topologie à switch (exercice 2) car elle est la plus complète. Les adresses sont celles du plan d'adressage ci-dessus ; les tests portent cette fois sur des trajets différents de l'exercice 2.

```
Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

```
Pinging 192.168.1.14 with 32 bytes of data:

Reply from 192.168.1.14: bytes=32 time=1ms TTL=128
Reply from 192.168.1.14: bytes=32 time<1ms TTL=128
Reply from 192.168.1.14: bytes=32 time<1ms TTL=128
Reply from 192.168.1.14: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.14:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

```
Reply from 192.168.1.16: bytes=32 time<1ms TTL=128
Reply from 192.168.1.16: bytes=32 time<1ms TTL=128
Reply from 192.168.1.16: bytes=32 time<1ms TTL=128
Reply from 192.168.1.16: bytes=32 time<1ms TTL=128

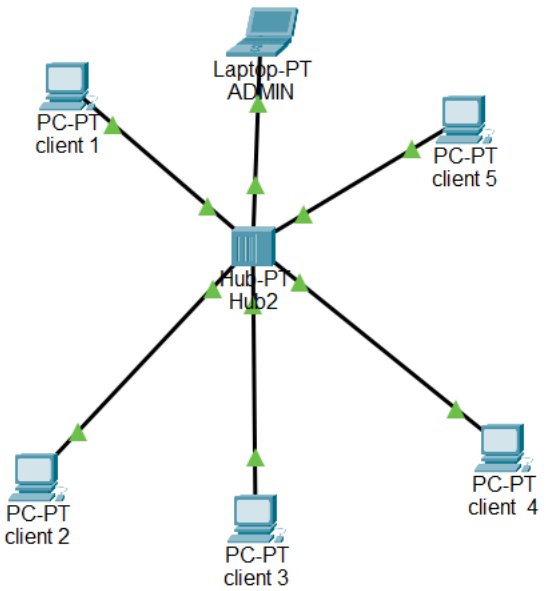
Ping statistics for 192.168.1.16:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

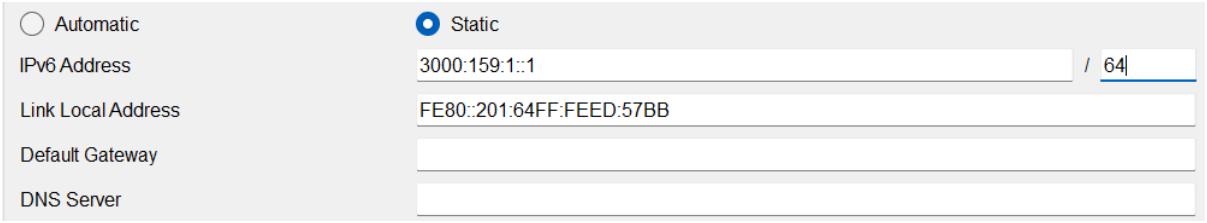
Server Mail joint les clients et Admin avec 0 % de perte. La connectivité est donc effective

Exercice 4 – Topologie à hub, adresses IPv6

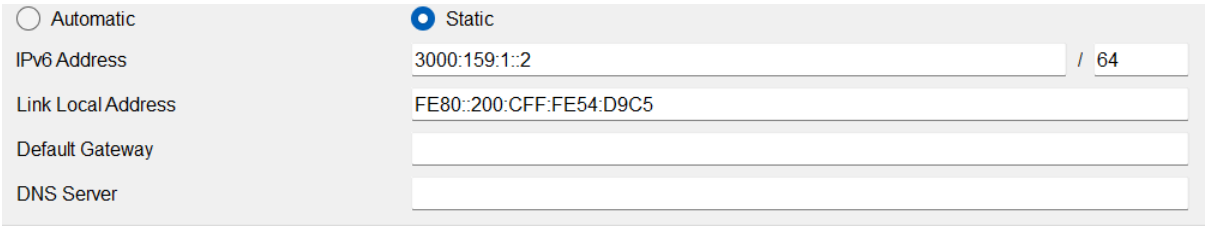
Appareil	Adresse IPv6	Préfixe
Client1	3000:159:1::1	/64
Client2	3000:159:1::2	/64
Client3	3000:159:1::3	/64
Client4	3000:159:1::4	/64
Client5	3000:159:1::5	/64
Admin (Laptop-PT)	3000:159:1::6	/64

Chaque appareil reçoit son adresse IPv6 avec le préfixe 64, dans le réseau .

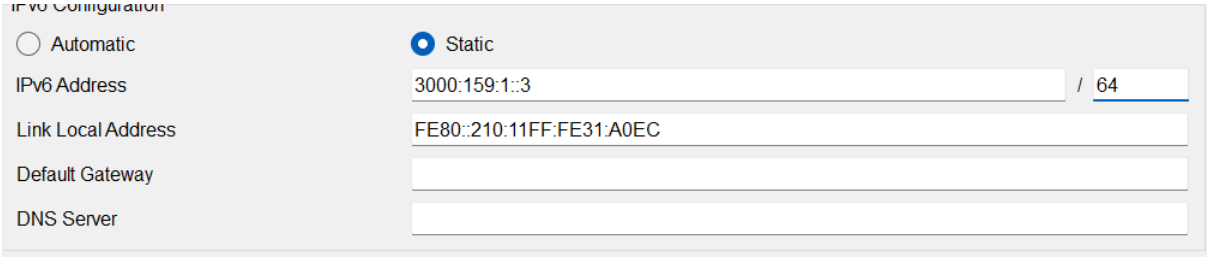




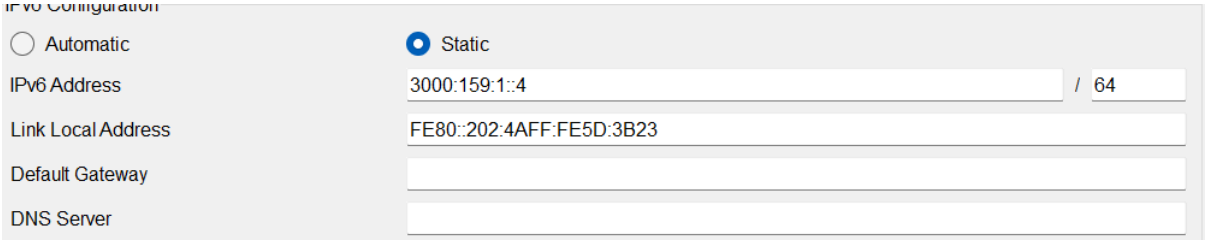
Client 1



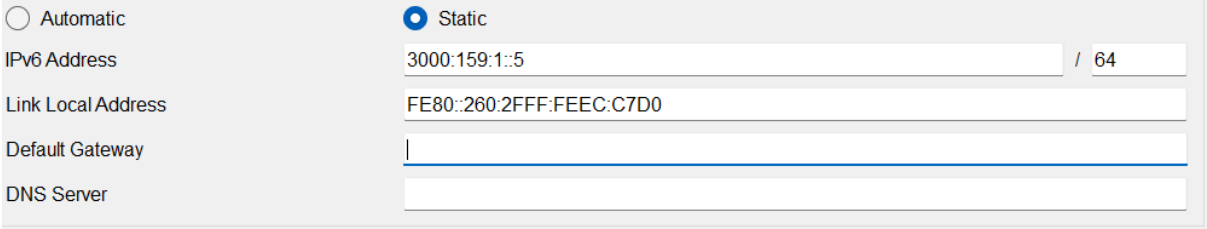
Client 2



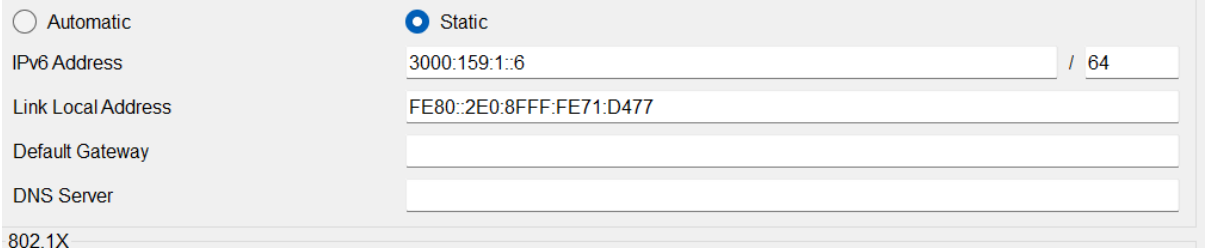
Client 3



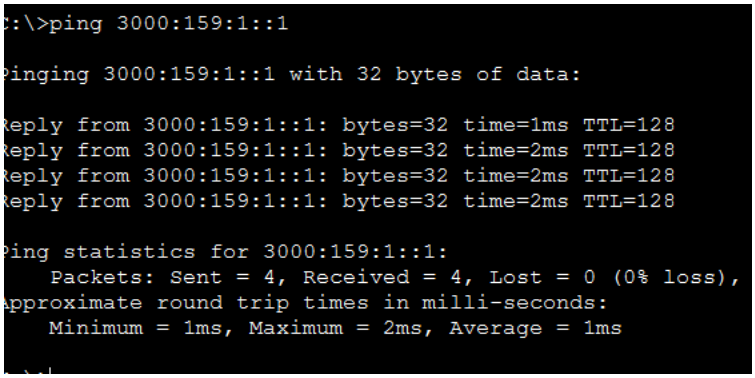
Client 4

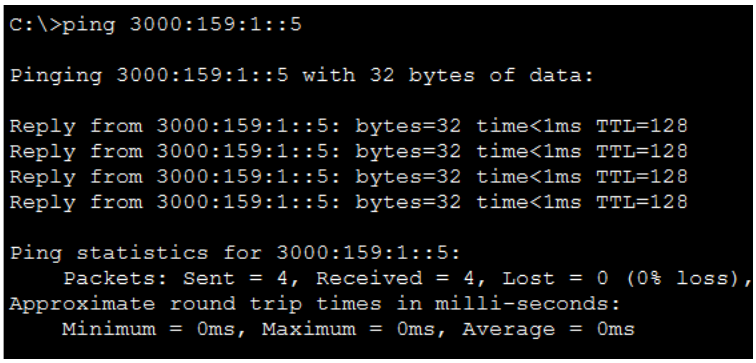


Client 5



Laptop-Admin

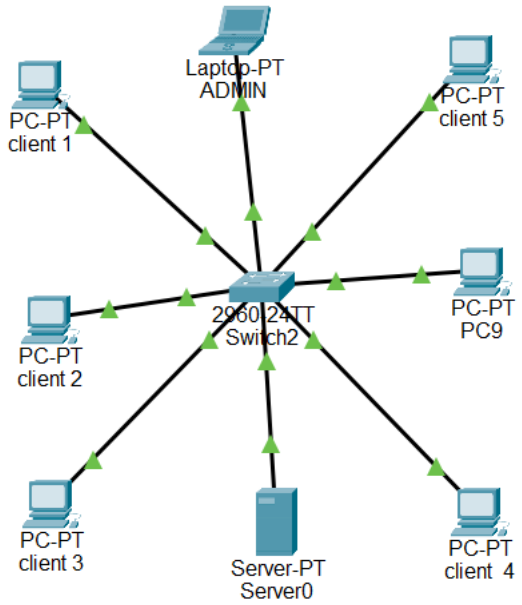




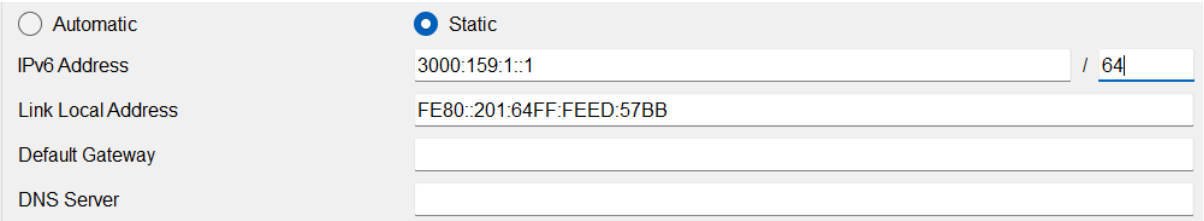
Les réponses reçues montrent que les appareils communiquent en IPv6 à travers le hub.

Exercice 5 – Topologie à switch, adresses IPv6

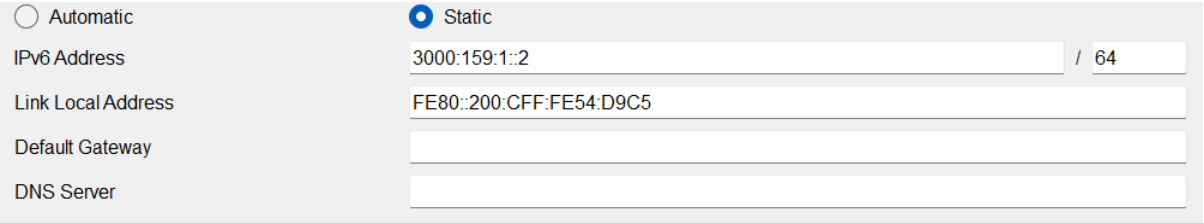
Appareil	Adresse IPv6	Préfixe
Client1	3000:159:1::1	/64
Client2	3000:159:1::2	/64
Client3	3000:159:1::3	/64
Client4	3000:159:1::4	/64
Client5	3000:159:1::5	/64
Client6	3000:159:1::6	/64
Admin (Laptop-PT)	3000:159:1::7	/64
Server Mail (Server-PT)	3000:159:1::8	/64



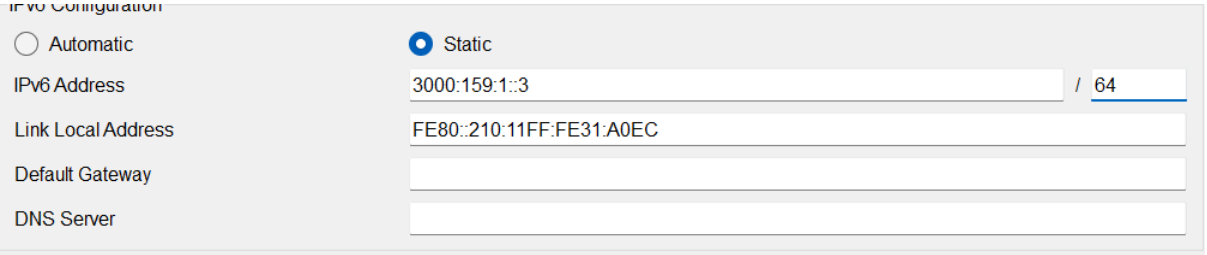
Les huit appareils sont configurés avec les adresses 3000:159:1::1 à 3000:159:1::8, préfixe /64.



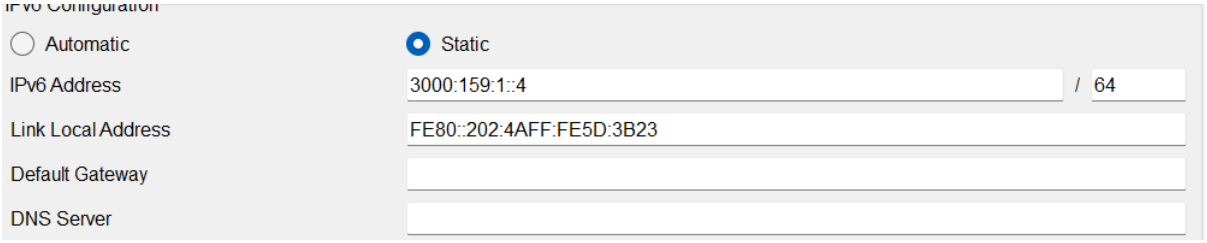
Client 1



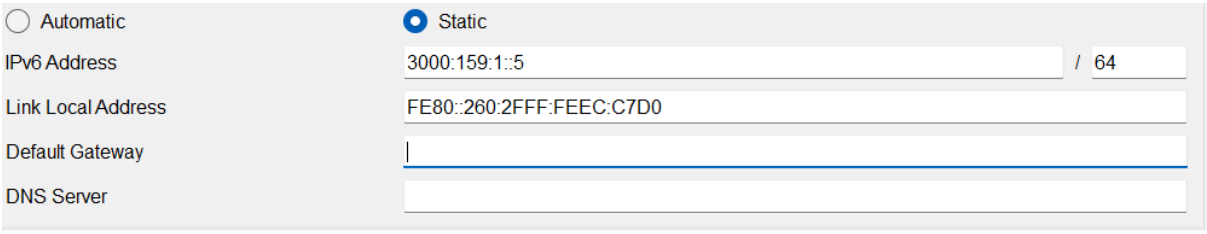
Client 2



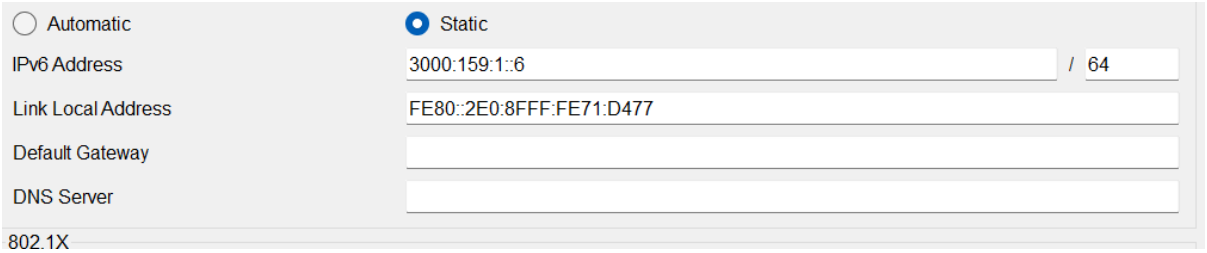
Client3



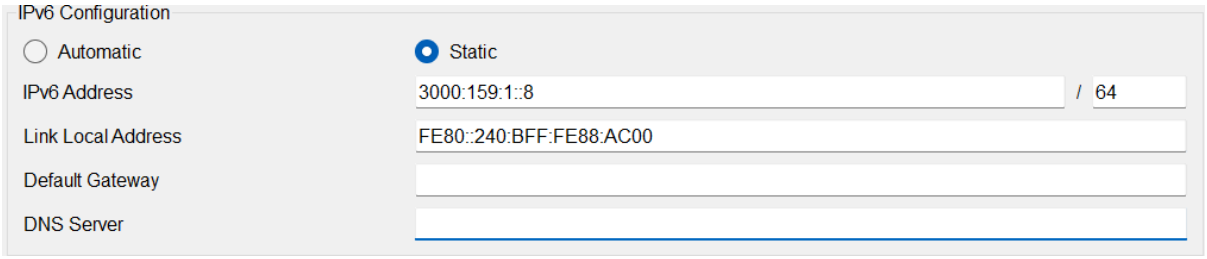
Client4



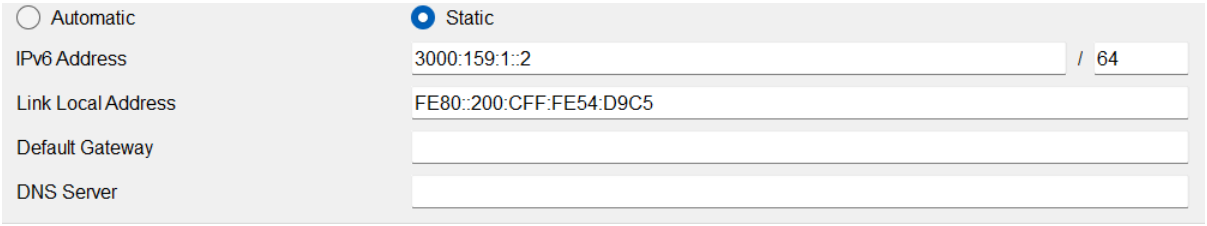
Client 5



Client 6



Laptop Admin



Serveur

```
C:\>ping 3000:159:1::8

Pinging 3000:159:1::8 with 32 bytes of data:

Reply from 3000:159:1::8: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::8: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::8: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::8: bytes=32 time<1ms TTL=128

Ping statistics for 3000:159:1::8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>ping 3000:159:1::7

Pinging 3000:159:1::7 with 32 bytes of data:

Reply from 3000:159:1::7: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::7: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::7: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::7: bytes=32 time<1ms TTL=128

Ping statistics for 3000:159:1::7:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Client1 reçoit les réponses de Server Mail et d'Admin : la connectivité IPv6 est établie à travers le switch.

3. Conclusions

3.1 Ce que j’ai appris

- Une adresse IPv4 valide dépend de son masque : les hôtes du même réseau doivent partager le même préfixe
- Une adresse IPv6 comporte 128 bits, s’écrit en hexadécimal et se simplifie avec « :: ». Avec le préfixe /64, la partie réseau est de 64 bits.
- La configuration d’un hôte dans Packet Tracer se fait par Desktop puis IP Configuration (ou Config pour un serveur).
-